

From Stars to Statistics: Understanding Globular Clusters with Data

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Topics we will cover today:

- **Gaussian Distributions:** The bell curve and how to describe it
- **Likelihood:** How to measure whether data supports a hypothesis
- **Mixture Models:** How to unmix overlapping populations
- **Bayes' Theorem:** How to combine data with prior knowledge
- **MCMC Sampling:** How to explore the space of possible solutions



Link to the notebook:

<https://colab.research.google.com/drive/1UfYfpl7t8fVzLu0c4-6m191DYB2YfTKU?usp=sharing>

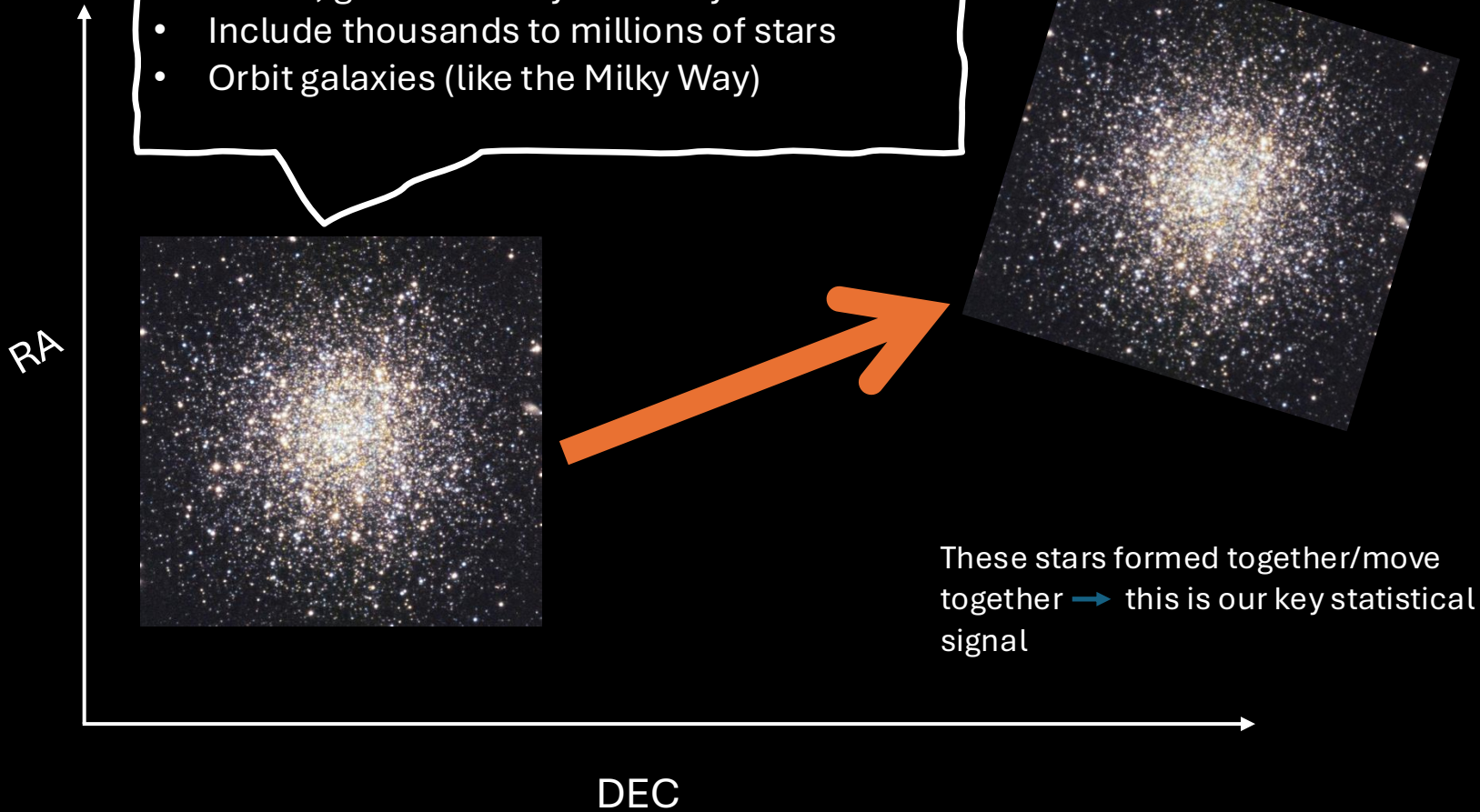
The story of NGC 288, a “Globular Cluster”

- Dense, gravitationally bound system of stars
- Include thousands to millions of stars
- Orbit galaxies (like the Milky Way)



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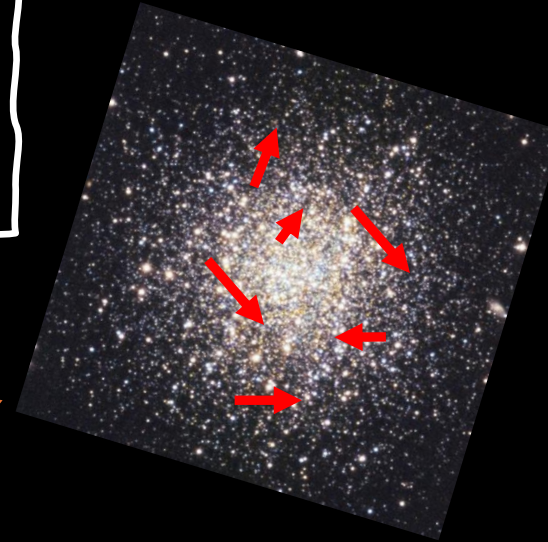


proper motion: the apparent movement of a star across the sky over time (measured in milliarcseconds per year, or mas/yr).

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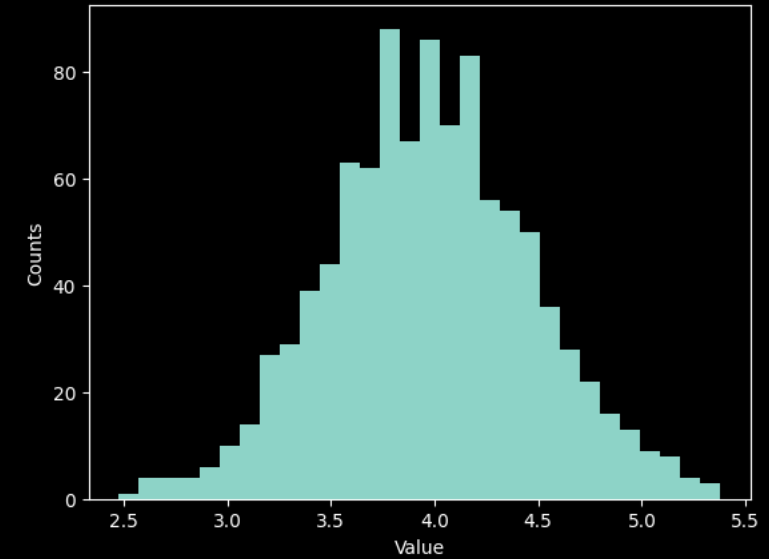
RA



These stars formed together/move together → this is our key statistical signal

DEC

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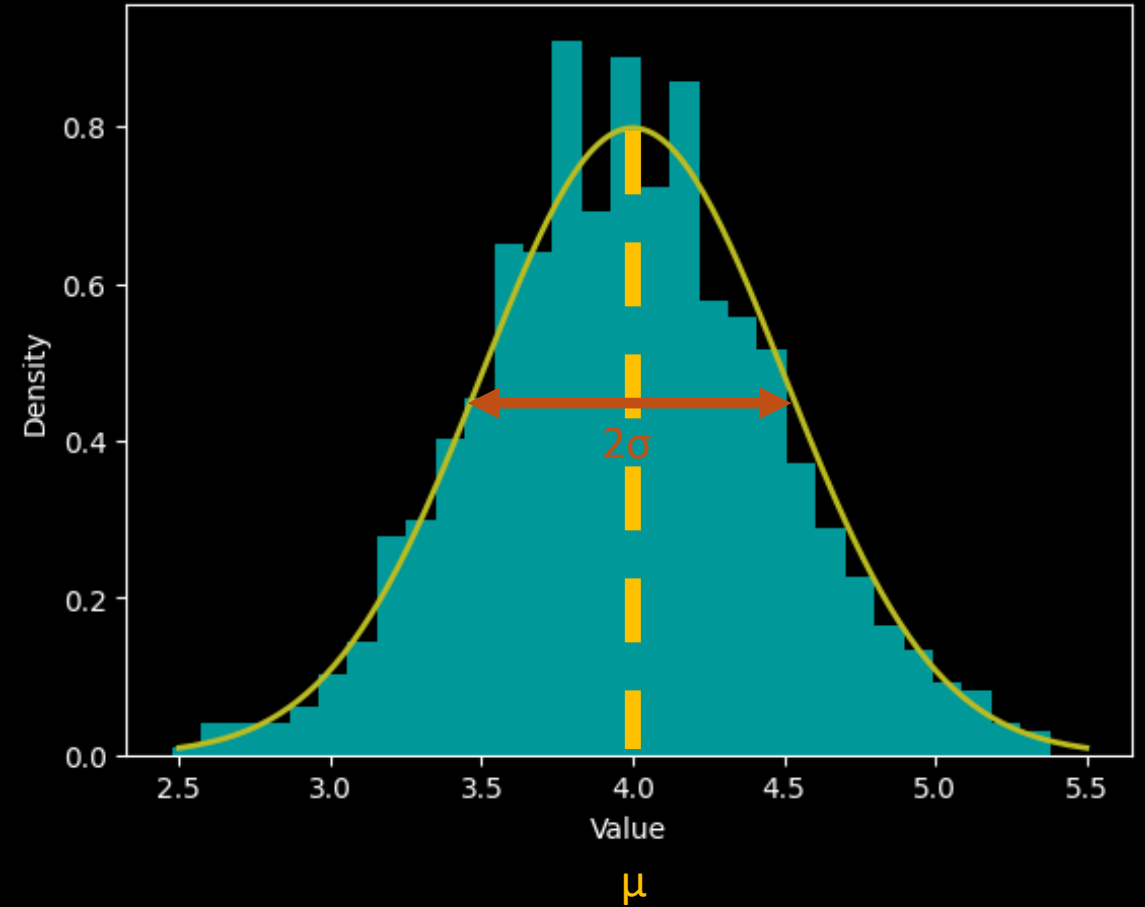
Proper motion values for each star

Measurement noise + small intrinsic dispersion → spread

Gaussian Distribution

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

- Mean (μ): center of the distribution
- Standard deviation (σ): width of the distribution



Likelihood

Likelihood(θ) = $P(D \mid \theta)$ = probability of observing data D if θ is true

Explain simply:

“How well does a model explain the data?”

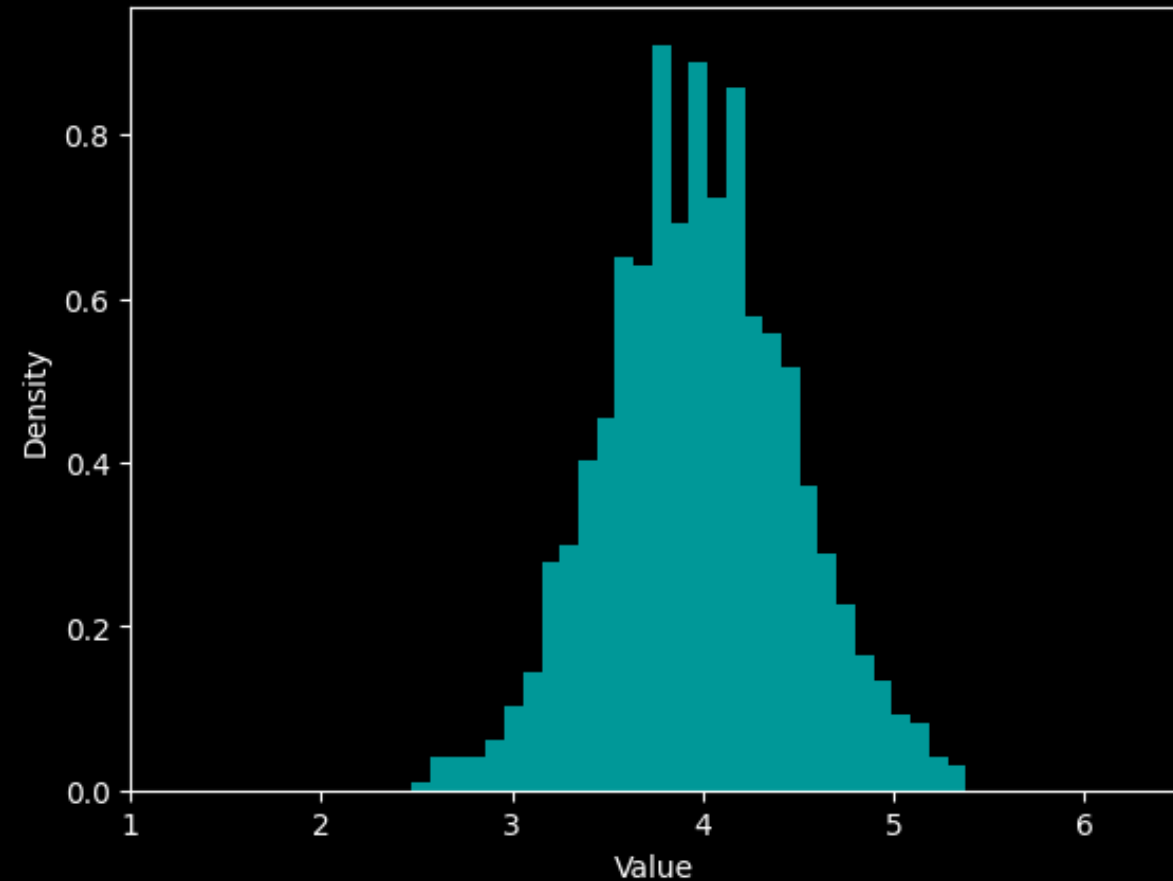
For data points x_i :

$$\mathcal{L} = \prod_{i=1}^N P(x_i)$$

👉 we maximize this

We usually work with $\log(\text{likelihood})$ for practical reasons:

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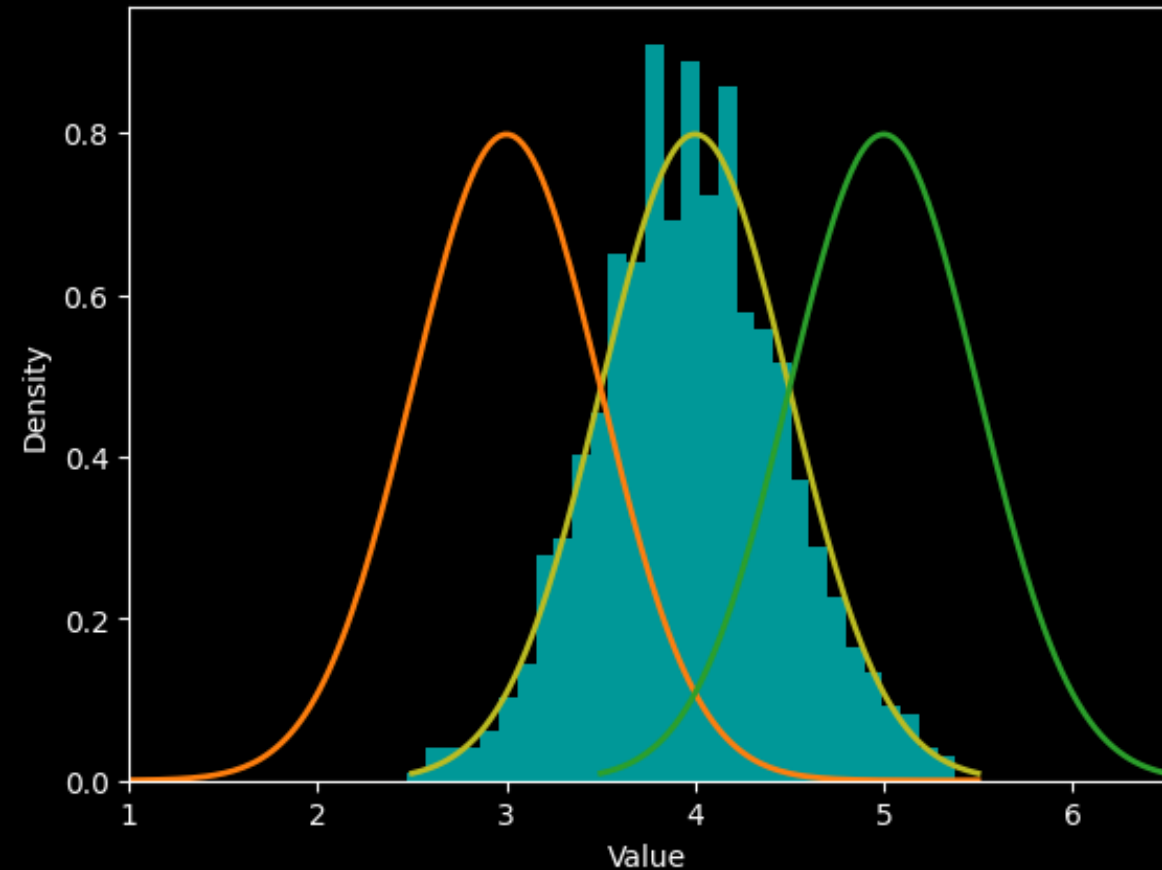
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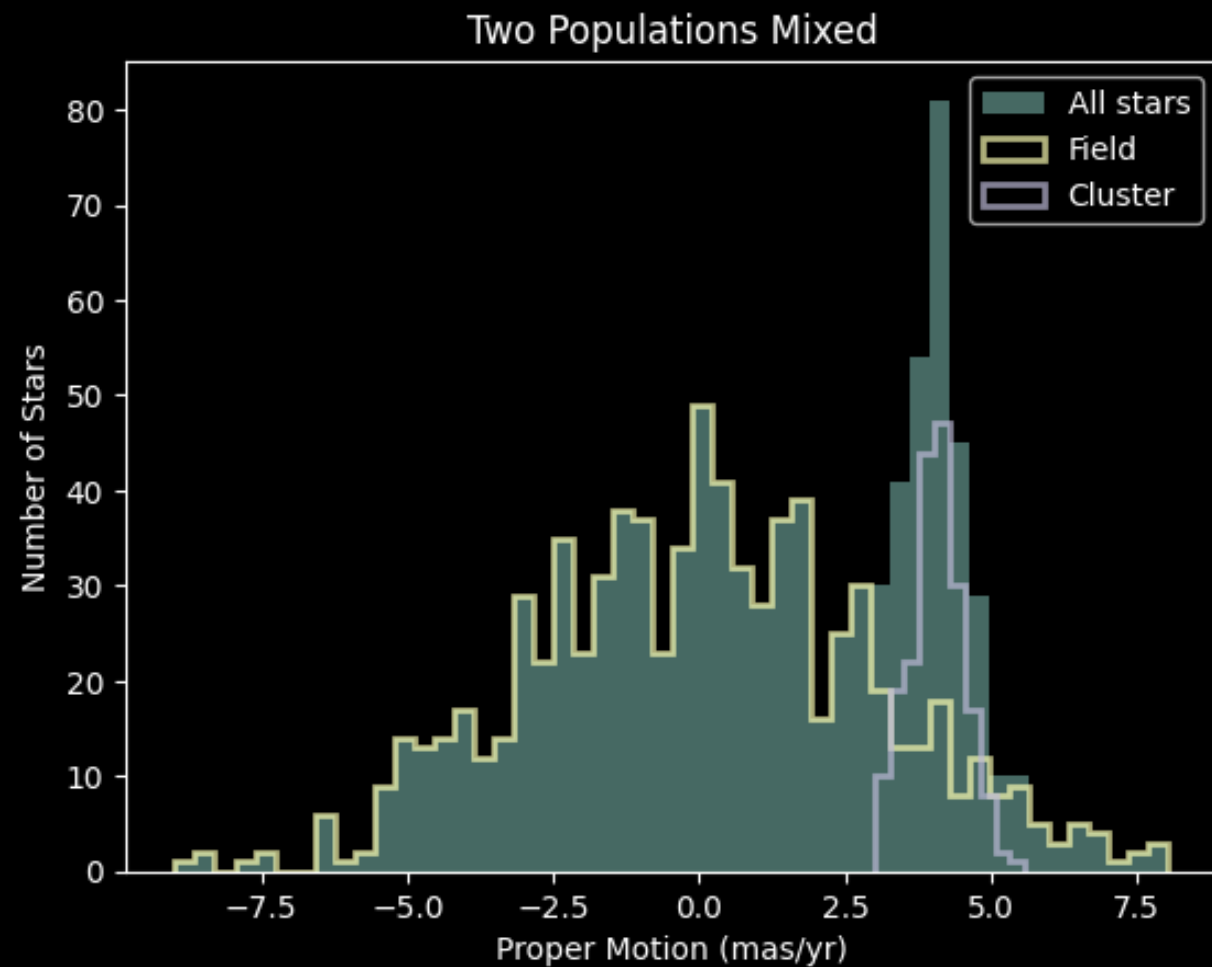
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Two Populations/Mixture Models



Likelihood for our Mixture Model:

Simplistically we can write:

Total = cluster + background

Then formalize:

$$P(x) = f \cdot \underbrace{\mathcal{N}(x \mid \mu_c, \sigma_c^2)}_{\substack{\text{Normal distribution} \\ \text{of cluster members}}} + (1 - f) \cdot \underbrace{\mathcal{N}(x \mid \mu_f, \sigma_f^2)}_{\substack{\text{Normal distribution} \\ \text{of field members}}}$$

f Fraction of cluster stars

Now same as before we calculate log(likelihood):

$$\log(\mathcal{L}) = \sum_{i=1}^N \log(P(x_i)) \quad \text{👉 then we maximize it}$$

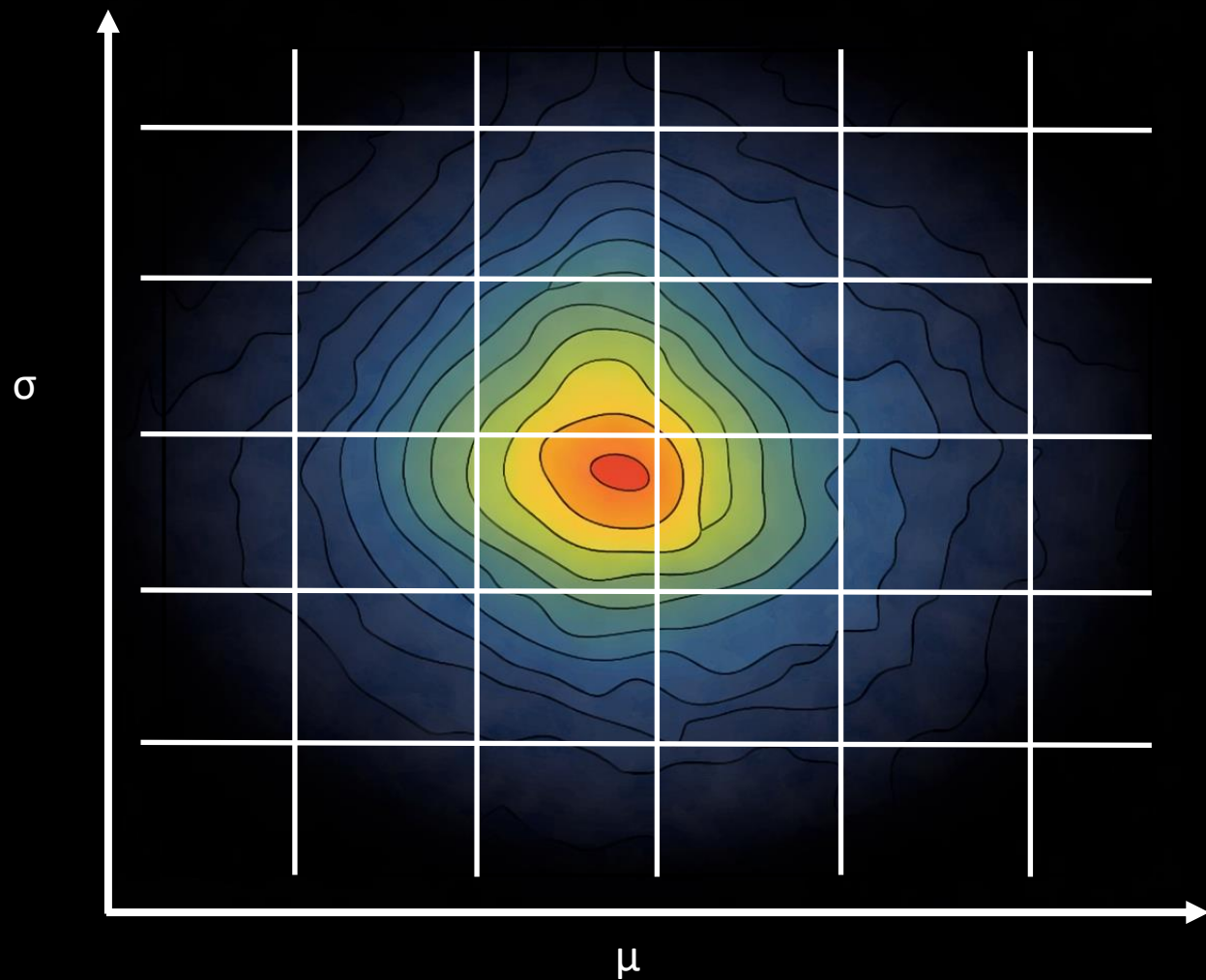
Bayes' Theorem

Conceptually:

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$

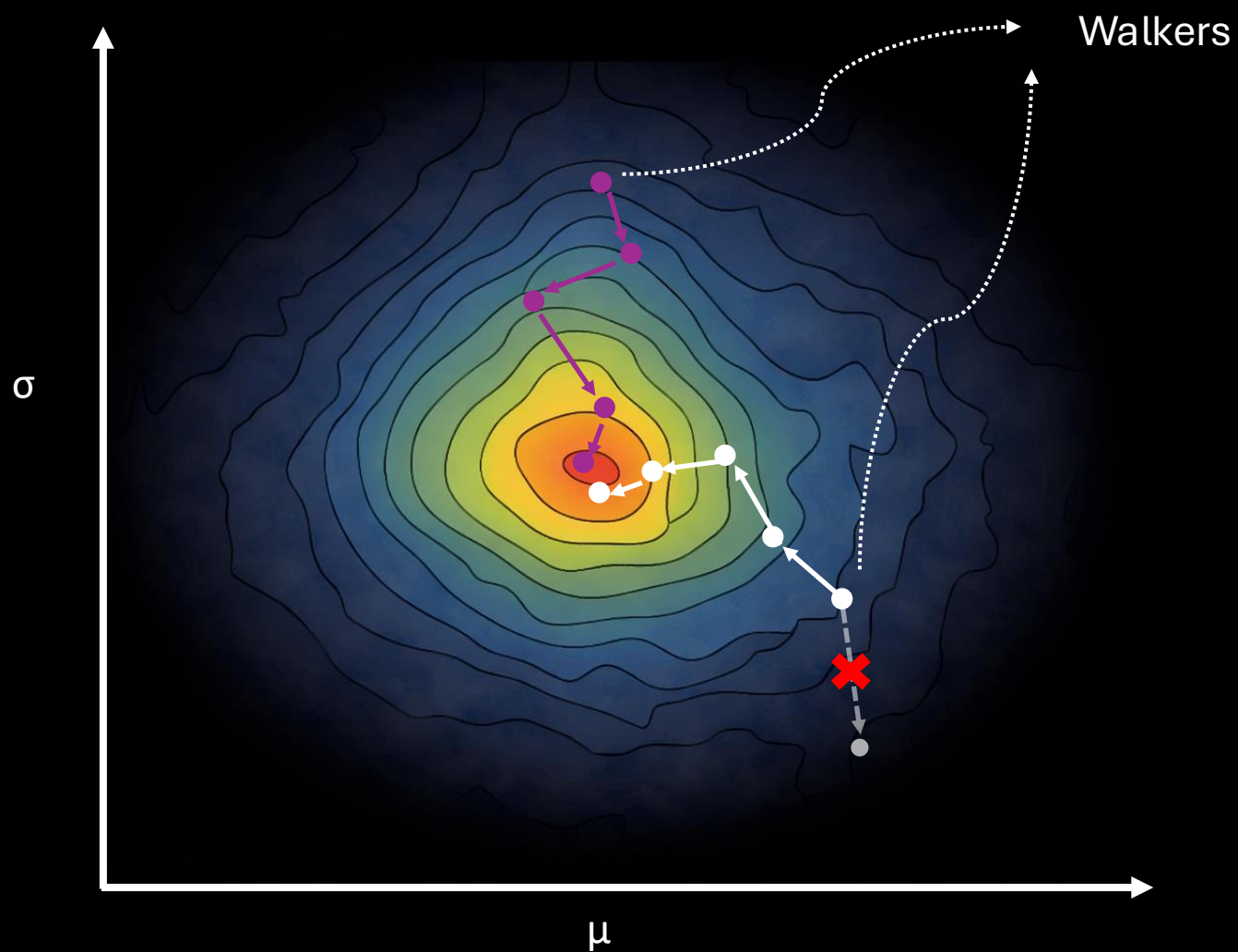
- Likelihood $\rightarrow$ data
- Prior $\rightarrow$ what we believe before
- Posterior $\rightarrow$ updated belief

Markov Chain Monte Carlo (MCMC):



Before: Using grid base method

Markov Chain Monte Carlo (MCMC):



Now: Walking randomly but preferring better regions